

Bias in Citizen Science Data: Implications for Mammalian Species Distribution Models

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Abstract

Citizen science (CS) platforms such as iNaturalist have become increasingly important sources of occurrence data for mammalian species distribution (SDMs), particularly in regions where structured survey data are limited. Despite their growing value for conservation and ecological research, these datasets are affected by substantial spatial, temporal, taxonomic, and detectability biases that can distort model outputs and reduce predictive reliability. These challenges are especially pronounced in mammals, many of which are nocturnal, cryptic, wide-ranging, or otherwise difficult to detect through opportunistic observation alone. This review synthesizes current literature examining how observer-driven biases influence mammalian SDMs derived from citizen science data and evaluates the efficacy of commonly used correction strategies. Spatial clustering near roads, trails, and urban centers consistently emerged as one of the strongest sources of bias, while uneven observer effort across seasons and preferential reporting of large-bodied or charismatic species further altered inferred habitat suitability and distribution patterns.

This review also highlights important differences between mammalian and avian CS datasets, particularly in the availability of effort metadata and the ability to account for imperfect detection. Approaches such as spatial thinning, target-group background selection, effort covariates, occupancy modeling, and the integration of structured survey data with opportunistic records have improved model performance, although no single method fully resolves the limitations associated with unstructured datasets. As citizen science continues to expand within wildlife research, the development of bias-aware and detection-informed modeling frameworks will be essential for improving

the reliability of mammalian SDMs and supporting evidence-based conservation planning.

Keywords

Citizen science; habitat modeling; MaxEnt; mammals; sampling bias; species distribution models

Introduction

Citizen science (CS) platforms have become important tools for biodiversity monitoring and ecological research, and for mammals, which are often more cryptic and wide-ranging, this has become an increasingly important resource for habitat modeling ([Dickinson, Zuckerberg, and Bonter, 2010](#); [Tye et al., 2017](#)). Platforms such as iNaturalist now provide millions of georeferenced observations that are routinely incorporated into ecological research, conservation planning, and species distribution modeling. For mammals, these datasets offer substantial potential because many species are difficult to survey systematically across large spatial scales due to their wide-ranging behavior, low population densities, or cryptic activity patterns. Opportunistic observations collected by the public can therefore provide valuable supplementary information on species occurrence, range expansion, and habitat use. Despite these advantages, CS data introduces methodological challenges that differ fundamentally from those associated with structured ecological surveys. Observations submitted to community science platforms are shaped by observer behavior rather than standardized sampling protocols, producing uneven patterns in where, when, and which species are recorded ([Milanesi, Mori, and Menchetti, 2020](#)). As a result, citizen science

datasets frequently contain strong spatial, temporal, and taxonomic bias that can distort ecological inference if left uncorrected. Many of these biases are particularly pronounced in mammals. Compared with birds and conspicuous taxa, mammals are often more difficult to detect because many species are nocturnal, elusive, sparsely distributed, or active in densely vegetated habitats. Mammalian observations are therefore more strongly influenced by imperfect detection and observer accessibility, increasing the likelihood that recorded distributions reflect patterns of human activity rather than true species patterns. In addition, mammal-focused citizen science data are commonly derived from unstructured platforms such as iNaturalist, where information on search effort and non-detections is limited or absent. These constraints complicate the use of conventional SDM approaches and make bias correction more difficult than in semi-structured systems such as eBird.

Recent advances in bias-aware modeling have improved researchers' ability to account for these limitations. Approaches including spatial thinning, target-group background selection, occupancy modeling, effort covariates, and the integration of CS data with structured surveys have all shown promise in reducing sampling bias and improving predictive performance. However, the effectiveness of these methods varies substantially depending on data quality, species ecology, and the extent of observer-driven bias present within a given dataset.

This review synthesizes current research on how systematic biases in citizen science data influence mammalian species distribution models. Particular attention is given to spatial, temporal, taxonomic, and detectability-related biases, as well as the methodological strategies developed to address them. By focusing on mammals, this

synthesis highlights challenges that are often underrepresented in broader CS literature and examines the implications of these biases for ecological inference and evidence-based conservation planning.

Methods

This review synthesizes recent literature examining how systematic biases in citizen science data influence mammalian species distribution models. The primary objective was to evaluate the major forms of bias associated with opportunistic biodiversity observations and to examine the methodological approaches used to reduce their effects on ecological inference and predictive modeling.

Relevant literature was identified through targeted searches of peer-reviewed ecological and conservation science publications between 2010 and 2025, a period corresponding with the rapid expansion of digital CS platforms such as iNaturalist. Searches focused on studies involving citizen science data, species distribution modeling, and bias correction methods relevant to mammalian ecology. Search terms included combinations of keywords such as “*citizen science*,” “*iNaturalist*,” “*species distribution model*,” “*SDM*,” “*mammals*,” “*sampling bias*,” “*spatial bias*,” “*detectability*,” “*occupancy modeling*,” “*MaxEnt*,” “*pseudo-absence*,” and “*observer effort*.”

Additional studies were identified through reference tracing from influential review papers and frequently cited methodological studies in the SDM literature. Both empirical and methodological studies were considered when they provided insight into the effects of spatial, temporal, taxonomic, or detection-related bias in opportunistic biodiversity datasets. Although the review focuses primarily on mammals, selected avian and

broader biodiversity datasets were also included when they addressed methodological developments widely applied across taxa.

The literature was evaluated qualitatively to identify recurring sources of bias, common modeling approaches, and emerging strategies for improving SDM performance when using unstructured CS data. Particular attention was given to studies examining imperfect detection, observer-driven sampling patterns, spatial filtering approaches, occupancy modeling frameworks, and the integration of citizen science observations with structured survey data.

Results

Across the reviewed literature, citizen science datasets consistently exhibited non-random sampling patterns that influenced the performance and reliability of mammalian species distribution models. Spatial bias emerged as the most persistent and influential source of error, although temporal variation, taxonomic selectivity, and imperfect detection also contributed substantially to uncertainty in predicted distributions. Because opportunistic observations are shaped by observer behavior rather than standardized survey design, many SDMs derived from citizen science data risk modeling patterns of human accessibility and reporting effort rather than true mammalian species occurrence.

The reviewed studies demonstrated that the influence of bias varied across modeling frameworks and taxonomic groups. Presence-only approaches, such as MaxEnt, remained the most commonly used tools for analyzing unstructured CS data because of their flexibility and ability to accommodate sparse occurrence records. However, these

models were also highly sensitive to uneven sampling effort and spatial clustering when bias correction methods were not incorporated.

Spatial bias was identified as the most common source of distortion in CS-derived SDMs. Mammalian observations were frequently concentrated near roads, trails, urban centers, and recreational areas where observers were more likely to encounter and document wildlife ([Cretois et al., 2021](#); [Geurts et al., 2023](#)). This accessibility bias often produced inflated estimates of habitat suitability in human-modified landscapes while underrepresenting remote or less accessible habitats. In several studies, the spatial distribution of observations more closely reflected observer movement patterns than underlying species ecology.

Road-associated clustering was especially prominent in studies using iNaturalist and other opportunistic biodiversity platforms. Observations of mammals, including squirrels and large ungulates, were disproportionately recorded near transportation corridors and areas of high human activity ([Boyat and Arévalo-Huezo, 2025](#); [Tye et al., 2017](#)). These patterns can lead SDMs to infer false associations between mammalian species patterns and anthropogenic environments, particularly when background selection does not adequately account for uneven observer effort ([Bonnet-Lebrun et al., 2018](#); [Sierra et al., 2025](#)). Spatial autocorrelation further amplified these issues by increasing redundancy among nearby occurrence records and artificially inflating model performance metrics. In undersampled regions, sparse occurrence records often restricted SDMs to well-surveyed landscapes, reducing the ability of models to accurately extrapolate into less accessible habitats. This limitation can ultimately affect

conservation planning by underestimating suitable habitat outside areas of concentrated human activity.

Temporal biases also influenced the structure and reliability of mammalian CS datasets. Observation effort frequently increased during weekends, holidays, and periods of favorable weather when recreational activity was highest ([Callaghan et al., 2021](#); [Sierra et al., 2025](#)). Seasonal fluctuations in observer effort integrated with mammalian behavior to produce uneven detection patterns throughout the year. Species exhibiting hibernation, migration, or seasonal activity shifts were particularly vulnerable to these effects, as periods of reduced detectability could be misinterpreted as ecological absence rather than sampling limitation.

Long-term changes in platform participation introduced an additional source of temporal heterogeneity. The rapid growth of citizen science platforms over time increased observation density in more recent years, complicating attempts to infer temporal trends in mammalian species occurrence without accounting for changes in observer effort and reporting intensity.

Taxonomic bias further distorted mammalian occurrence datasets by favoring species that were more conspicuous, charismatic, or easily photographed ([Drury et al., 2019](#)). Large-bodied mammals and visually recognizable carnivores were disproportionately represented in CS platforms, whereas small, cryptic, nocturnal, or fossorial species were consistently underreported ([Callaghan et al., 2021](#)). Observer preferences and public interest, therefore, shaped which taxa were most frequently documented, producing uneven representation across mammalian communities.

Many methodological advances in citizen science SDMs have been developed using avian datasets, particularly through semi-structured platforms such as eBird. However, the reviewed literature suggests that these approaches do not always transfer effectively to mammalian systems because mammals present distinct detectability challenges and data limitations.

Compared with birds, mammals are often more difficult to observe consistently due to nocturnal behavior, lower population density, cryptic coloration, and reduced vocal detectability. Many mammalian species also occupy densely vegetated habitats or exhibit avoidance behavior in response to human presence, further limiting detection opportunities. Consequently, non-detection in mammalian datasets frequently reflects imperfect observation rather than true absence. These challenges become especially prevalent for rare or infrequently detected species, where limited sighting frequency can substantially reduce SDM reliability and increase uncertainty in predicted distributions ([Virgili et al., 2018](#)).

Differences in data structure further compound these challenges. Avian CS datasets commonly include semi-structured checklists containing metadata on sampling effort, including survey duration, travel distance, and observer number ([Johnston et al., 2021](#); [Kelling et al., 2019](#)). These data allow researchers to model detection probability more explicitly and apply occupancy-based approaches with greater confidence ([Peterman, Crawford, and Kuhns, 2013](#)). In contrast, mammalian observations are more frequently derived from unstructured platforms such as iNaturalist, where records typically lack standardized information on search effort or survey conditions.

The absence of effort metadata limits the ability of researchers to distinguish ecological patterns from observer-driven processes. Consequently, mammalian SDMs are often more sensitive to detectability bias, uneven sampling effort, and false absences than comparable avian models. The reviewed studies consistently emphasized that approaches developed for birds may require substantial modification before being applied reliably to mammalian datasets.

A range of methodological approaches has been developed to reduce the effects of bias in citizen science-derived SDMs. Spatial filtering and thinning were among the most widely used correction methods, reducing the influence of clustered occurrence records by removing observations located within a specified distance threshold ([Ruzzier et al., 2024](#)). These approaches improved model stability by decreasing spatial autocorrelation and limiting the overrepresentation of heavily surveyed locations.

Background manipulation and target-group background selection were also commonly employed to address uneven sampling effort. Rather than selecting pseudo-absences randomly, these approaches draw background points from areas known to have been surveyed by observers, thereby aligning pseudo-absence selection with underlying sampling processes ([Botella et al., 2020](#); [Milanesi, Mori, and Menchetti, 2020](#)). This strategy reduces the likelihood that models interpret unsampled locations as true absences.

Several studies incorporated proxies for observer effort directly into SDMs. Variables such as observation density, number of observers, or total records within a grid cell were used to account for spatial and temporal variation in sampling intensity ([Milanesi, Mori, and Menchetti, 2020](#)). However, the literature also highlighted a persistent

limitation of mammalian CS datasets: the absence of standardized effort metrics comparable to those available in semi-structured avian monitoring systems.

The integration of citizen science observations with structured ecological datasets emerged as one of the most effective strategies for improving model reliability.

Combining opportunistic records with telemetry, camera trap surveys, hunting records, or systematic field observations allowed researchers to offset the spatial noise and detection limitations associated with unstructured CS data ([Feldman et al., 2021](#); [Rutten et al., 2019](#)). Data integration approaches were particularly valuable in highly modified landscapes where observer bias was strongest. Similar benefits have also been observed when opportunistic biodiversity photographs were used to complement structured ecological surveys ([Roberts et al., 2022](#)).

Recent advances in occupancy modeling and Bayesian hierarchical approaches have further improved researchers' ability to account for imperfect detection and undersampling. Unlike conventional presence-only SDMs, occupancy models explicitly estimate both occurrence probability and detection probability, allowing researchers to distinguish imperfect detection from ecological absence more effectively ([Guillera-Arroita, Kéry, and Lahoz-Monfort, 2019](#)). Bayesian implementations have proven especially useful for sparse mammalian datasets because they improve inference under conditions of limited sampling and uneven effort ([Outhwaite et al., 2018](#)). Additional emerging approaches, including sequential estimation methods and uncertainty-based modeling frameworks, have also shown promise for improving predictions derived from opportunistic biodiversity observations ([Mengersen et al., 2017](#); [Ver Hoef et al., 2021](#)).

Discussion

The growing availability of citizen science data has substantially expanded opportunities for mammalian ecological research and species distribution modeling. Platforms such as iNaturalist provide occurrence records across geographic regions and time periods that would be difficult to sample consistently through traditional field surveys alone. For many mammals, particularly wide-ranging or poorly monitored species, these observations represent an increasingly important source of ecological information. However, the reviewed literature demonstrates that opportunistic biodiversity datasets are shaped strongly by observer behavior, uneven accessibility, and imperfect detection, all of which can introduce substantial noise into ecological inference if not properly addressed ([Isaac et al., 2014](#)).

Among the various forms of bias identified, spatial bias consistently emerged as the most influential source of model distortion. Mammalian observations were concentrated disproportionately near roads, trails, urban centers, and recreational areas where human activity and observer access were highest. As a result, SDMs risk overestimating habitat suitability in accessible landscapes while underrepresenting remote or less frequently surveyed habitats. These findings reinforce concerns that uncorrected citizen science datasets may reflect patterns of observer movement as much as underlying species ecology. The effects of this bias are particularly important in conservation planning, where inaccurate habitat predictions may influence management decisions, protected area prioritization, and assessments of species vulnerability.

The literature also highlights the importance of imperfect detection as a defining challenge in mammalian citizen science research. Compared with birds and other highly

detectable taxa, mammals are often more difficult to observe consistently because many species are nocturnal, elusive, sparsely distributed, or behaviorally cryptic. Non-detection in mammalian datasets, therefore, frequently reflects limited detectability rather than true absence. When detection probability varies systematically across habitats or regions, model outputs may become biased even when large numbers of observations are available. The magnitude of these effects may also depend on species ecology, as observational noise and sampling bias can disproportionately affect habitat specialists and narrowly distributed taxa ([Soultan and Safi, 2017](#)).

The reviewed studies indicate that no single correction method fully resolves the limitations associated with opportunistic biodiversity data. Instead, the most effective approaches generally combine multiple bias mitigation strategies tailored to the ecological characteristics of the focal species and the structure of the dataset. Spatial thinning and target-group background selection reduce the influence of clustered observations, while occupancy modeling and Bayesian hierarchical approaches improve the treatment of imperfect detection and sparse occurrence data. Similarly, integrating citizen science observations with structured ecological datasets, including camera trap surveys, telemetry data, and systematic field sampling, consistently improved model reliability and predictive performance.

The increasing use of integrated modeling frameworks may represent one of the most important directions for future mammalian SDM research ([Tuia et al., 2024](#)). Combining opportunistic citizen science observations with independent survey methods allows researchers to capitalize on the broad spatial coverage of public biodiversity platforms while reducing uncertainty associated with uneven observer effort and detection

probability. These approaches may be especially valuable for rare, cryptic, or conservation-sensitive mammals that are difficult to monitor using any single survey method alone.

Several limitations of the present review should also be acknowledged. The available literature remains uneven across mammalian taxa and geographic regions, with most studies focusing on large-bodied or charismatic species and on regions with relatively high citizen science participation. In addition, rapid changes in platform participation and data volume may alter the performance of modeling approaches over time. Continued evaluation of emerging bias-correction methods will therefore be necessary as citizen science datasets become more prevalent and diverse.

Despite these limitations, citizen science platforms are likely to remain an increasingly important component of mammalian biodiversity monitoring. The continued growth of publicly accessible occurrence databases presents significant opportunities for conservation research, particularly in regions where traditional monitoring remains limited by cost or logistical constraints. However, the usefulness of these data ultimately depends on the ability of researchers to account for the observer-driven processes that shape opportunistic biodiversity records ([Arazy et al., 2024](#)). As citizen science datasets continue to expand, improving the treatment of imperfect detection and observer-driven bias will remain essential for ensuring that mammalian SDMs reflect ecological reality rather than patterns of human observation.

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Kailer S. Meredith (Conceptualization, Investigation, Writing–Original Draft, Writing–Review and Editing)

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Research involving Human Participants and/or Animals

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